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Ringraziamenti

...thanks to...

Contents

Abstract	2
1 Introduction and Background	3
1.1 Introduction	3
1.2 Research Question and Objectives	3
1.3 State of the Art	4
1.3.1 Deep Learning for Precision Agriculture	4
1.3.2 Remote Sensing and Sentinel-2 Time Series	5
1.3.3 Sequence Modeling: From CNNs to Transformers	5
1.3.4 Multi-Year Crop Rotation Modeling	6
1.4 Contributions	7
2 Methodology and System Design	8
2.1 The INVEKOS Dataset	8
2.1.1 Data Acquisition (Spatial Patches vs. 1D Pixel Data)	8
2.1.2 Pre-processing and Handling Irregular Time Series	8
2.2 Single-Year Non-Hybrid Baselines	9
2.2.1 Temporal 1D Convolutional Neural Network (CNN)	9
2.2.2 Temporal Gated Recurrent Unit (GRU RNN)	9
2.3 Multi-Year Hybrid Architecture	10
2.3.1 AgroDays Positional Encoding	10
2.3.2 The Hierarchical Transformer-LSTM Model	11
2.3.3 Capturing Long-Term Crop Rotations	12
2.4 Infrastructure and Optimization	13
2.4.1 PyTorch Lightning Training Pipeline	13
2.4.2 Distributed Hyperparameter Tuning with Ray Tune and Optuna	13
3 Experiments and Discussion	14
3.1 Experimental Setup	14
3.1.1 HPC Environment and I/O Bottleneck Mitigation	14
3.1.2 Data Loading Strategies	14
3.1.3 Evaluation Metrics	15
3.2 Baseline Results (Non-Hybrid Models)	15
3.2.1 Temporal CNN Performance	15
3.2.2 Temporal GRU RNN Performance	15
3.3 Proposed Architecture Results (Hybrid Model)	16
3.3.1 Hierarchical Transformer-LSTM Performance	16
3.4 Discussion	16
3.4.1 Single-Year vs. Multi-Year Capabilities	16
3.4.2 The Impact of Class Imbalance	17
4 Conclusions and Future Work	17
4.1 Conclusions	17
4.2 Future Work	17
Bibliography	17

Abstract

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Sommario è un breve riassunto del lavoro svolto dove si descrive l'obiettivo, l'oggetto della tesi, le metodologie e le tecniche usate, i dati elaborati e la spiegazione delle conclusioni alle quali siete arrivati.

Il sommario dell'elaborato consiste al massimo di 3 pagine e deve contenere le seguenti informazioni:

- contesto e motivazioni
- breve riassunto del problema affrontato
- tecniche utilizzate e/o sviluppate
- risultati raggiunti, sottolineando il contributo personale del laureando/a

1 Introduction and Background

1.1 Introduction

The global agricultural sector is currently facing unprecedented challenges. Driven by rapid population growth, climate change, and the need for sustainable resource management, modern agriculture must maximize crop yields while minimizing environmental impact [10]. Precision agriculture [20, 23] has emerged as a crucial solution to address these demands, relying heavily on accurate, large-scale, and up-to-date information regarding land use and crop distribution. Traditionally, acquiring this data required costly and time-consuming manual field surveys, which are impossible to scale effectively at a national or global level.

Over the past decade, Earth Observation (EO) satellites have revolutionized how we monitor agricultural lands. Constellations such as the European Space Agency’s Sentinel-2 provide free, high-resolution multispectral imagery with frequent revisit times (every 5 to 10 days) [7]. By observing how a specific parcel of land reflects light over time [32], it is possible to track the phenological cycle—the biological growth stages—of vegetation. This continuous stream of data allows researchers to identify specific crop types based on their unique growth signatures rather than relying on a single, static snapshot [35].

Simultaneously, the rise of Deep Learning (DL) has provided the computational tools necessary to process this massive influx of satellite data. Early approaches in remote sensing relied heavily on traditional machine learning algorithms and spatial pattern recognition. However, as the field evolved, it became evident that the temporal dimension—how a pixel changes over a year—holds the most predictive power for crop classification. Consequently, sequence-modeling architectures such as Temporal Convolutional Neural Networks (1D-CNNs) [16, 25] and Recurrent Neural Networks (RNNs) [21] have become the standard for agricultural time-series analysis.

Despite these advancements, operationalizing deep learning for crop mapping presents significant difficulties. First, Satellite Image Time Series (SITS) are inherently irregular; optical sensors cannot penetrate clouds, leading to missing data and uneven temporal gaps that disrupt traditional machine learning models. Second, processing high-dimensional spatial-temporal datasets—such as tracking entire 2D image patches over a temporal sequence—creates severe computational and memory bottlenecks, particularly in High-Performance Computing (HPC) environments. Finally, most current state-of-the-art models suffer from “temporal myopia”; they analyze only a single year of weather and phenology, completely ignoring the fundamental agricultural practice of crop rotation, where farmers intentionally alternate crops across multiple years to maintain soil health [4].

This thesis aims to bridge these gaps. To overcome the hardware bottlenecks associated with heavy spatial-temporal grids, this work shifts to a highly optimized 1D pixel-based pipeline. By analyzing the temporal evolution of individual pixels rather than broad spatial patches, the computational load is drastically reduced, enabling rapid prototyping and extensive hyperparameter optimization on an HPC cluster. Building upon this efficient infrastructure, the project establishes robust single-year temporal baselines—specifically 1D-CNNs and Gated Recurrent Units (GRUs)—and compares them against a novel, hybrid multi-year architecture: the Transformer-LSTM. By integrating the self-attention mechanism proven effective on satellite time series [33, 31] with the recurrent memory of an LSTM, the proposed system seeks to answer whether teaching a neural network the historical rules of multi-year crop rotation can significantly outperform models that rely on single-year phenology alone.

1.2 Research Question and Objectives

The goal of this thesis is to explore the efficacy of deep sequence-modeling architectures in classifying agricultural crop types using irregular SITS. Specifically, this research investigates the transition from traditional, single-year phenological analysis to a multi-year, hybrid approach capable of learning

long-term farming practices. This leads to the primary research question of this study:

“How can we efficiently process irregular Satellite Image Time Series for large-scale crop mapping, and does the integration of multi-year crop rotation history via a hybrid Transformer-LSTM architecture significantly outperform traditional single-year phenological models?”

To systematically answer this question, the project is structured around four specific objectives, spanning both machine learning infrastructure and architectural design:

- i) **Overcoming I/O and Computational Bottlenecks:** To redesign the data pipeline, shifting from heavy spatial-temporal patches to a highly optimized 1D pixel-based extraction method. This aims to resolve severe High-Performance Computing (HPC) memory constraints and reduce epoch training times, enabling rapid experimentation.
- ii) **Establishing Robust Single-Year Baselines:** To design, train, and tune standard non-hybrid temporal architectures; specifically a 1-Dimensional Convolutional Neural Network (1D-CNN) and a Gated Recurrent Unit Recurrent Neural Network (GRU-RNN). These models serve as the baseline for evaluating standard intra-year phenological classification.
- iii) **Developing a Multi-Year Hybrid Architecture:** To construct a novel Hierarchical Transformer-LSTM model that extends beyond a single harvest cycle. This objective focuses on using self-attention to manage intra-year weather irregularities and an LSTM memory cell to capture inter-year crop rotation dependencies.
- iv) **Automating Hyperparameter Optimization:** To build a robust, fully automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna. This ensures all models are evaluated fairly at their peak capacity through distributed hyperparameter tuning on an HPC cluster.

1.3 State of the Art

1.3.1 Deep Learning for Precision Agriculture

The paradigm of precision agriculture has fundamentally transformed modern farming management. By transitioning from uniform field-level treatments to highly granular, site-specific interventions, precision agriculture seeks to optimize resource inputs—such as water, fertilizers, and pesticides—while maximizing overall crop yield [20, 23]. This shift relies entirely on the continuous acquisition and analysis of vast amounts of heterogeneous data collected from on site Internet of Things (IoT) sensors, Unmanned Aerial Vehicles (UAVs), and Earth Observation (EO) satellites.

Historically, extracting actionable insights from these diverse data sources depended on traditional machine learning algorithms, such as Random Forests (RF) and Support Vector Machines (SVM) [22, 3]. While effective for specific, localized tasks, these traditional methods are inherently limited by their reliance on manual feature engineering. Researchers were required to construct and select specific vegetation indices or spatial textures beforehand, a process that is highly domain-specific, time-consuming, and difficult to scale across different geographical regions and climatic conditions.

The advent of Deep Learning (DL) has largely overcome these limitations, triggering a major shift in agricultural data analysis [14]. Unlike traditional machine learning, deep neural networks possess the capacity for automatic, hierarchical feature extraction. By ingesting raw or minimally processed data, DL models can autonomously learn complex, non-linear representations directly from the input space. In precision agriculture, this has led to unprecedented breakthroughs across a wide array of applications, including automated weed detection, early disease diagnosis, and precise yield estimation [15].

In the specific context of remote sensing and land cover mapping, deep learning has proven exceptionally proficient at handling the massive dimensionality of Satellite Image Time Series (SITS). By leveraging specialized architectures capable of modeling both spatial hierarchies and temporal dependencies, deep learning models can now accurately map crop distributions at a national scale, outperforming traditional shallow classifiers and forming the foundational technology upon which this thesis is built [17].

1.3.2 Remote Sensing and Sentinel-2 Time Series

The Importance of Spatio-Temporal Dynamics

The European Space Agency’s Sentinel-2 mission has revolutionized agricultural monitoring by providing high spatial resolution imagery with a frequent 5-day revisit time. This creates dense Satellite Image Time Series (SITS) that capture the vital phenological evolution of crops throughout the growing season [30]. However, extracting actionable insights from SITS requires models to navigate a complex time-space tradeoff. Research demonstrates that for crop type classification on Sentinel-2 data, the temporal structure is often richer than the spatial structure; consequently, hybrid models (like recurrent convolutional networks) perform best when the majority of their parameters are allocated to modeling temporal sequences rather than spatial features [30].

Encoding and Sequence Modeling

To effectively process these time series, researchers have explored various representation and sequence-modeling strategies:

- **Image-Based Encodings:** One approach involves transforming 1D time-series data (such as vegetation indices) into 2D image representations—like Gramian Angular Fields or Markov Transition Fields—allowing standard convolutional networks to efficiently extract static and dynamic temporal patterns [6].
- **Recurrent to Transformer Evolution:** While models like Long Short-Term Memory (LSTM) networks have traditionally handled sequential data, modern approaches increasingly rely on Transformer architectures to overcome LSTM limitations with long-range dependencies [37]. In crop mapping, advanced models like the Temporo-Spatial Vision Transformer (TSVIT) utilize purely self-attention mechanisms to process multi-modal SITS, combining optical data (Sentinel-2) with radar (Sentinel-1) to boost classification performance without relying on standard convolutions [9].

Overcoming Noise and Label Scarcity

Despite the rich information in SITS, operational mapping is frequently hindered by cloud cover noise and a severe lack of pixel-level ground truth data. Recent deep learning innovations address these bottlenecks directly:

- **Knowledge-Guided Networks:** To ensure models learn meaningful phenology rather than dataset artifacts, prior domain knowledge can be integrated into the architecture. For instance, the Knowledge-Guided Crop Mapping Network (KGCMNet) forces the model to reconstruct known vegetation indices (like NDVI) during training, which significantly sharpens its spatio-temporal representations [28].
- **Weakly Supervised Learning:** To bypass the need for exhaustive pixel-level annotations, weakly supervised frameworks (e.g., Exact) utilize only image-level labels. These models explore space-time perceptive clues to rectify erroneous semantic biases caused by anomalous temporal clips, pushing weakly supervised performance to nearly 95% of fully supervised baselines [38].
- **Multimodal Pseudo-Labeling:** Another highly scalable solution involves integrating top-down SITS with bottom-up, ground-level field images. By using multimodal data to generate reliable pseudo-labels, models can achieve high accuracy across vast geographical areas while drastically reducing the need for expensive on site data collection [1].

1.3.3 Sequence Modeling: From CNNs to Transformers

The accurate classification of agricultural land relies heavily on the ability to interpret the chronological sequence of crop phenology. As deep learning has evolved, so too have the architectures designed to model these complex temporal dynamics.

Convolutional Neural Networks (CNNs)

Early deep learning approaches for Satellite Image Time Series (SITS) relied heavily on Convolutional Neural Networks (CNNs). While originally designed to extract spatial features from static images [15], CNNs were successfully adapted for temporal sequences via 1-Dimensional (1D) convolutions. By

operating filters across the temporal dimension of a pixel’s profile, 1D-CNNs can efficiently capture localized temporal patterns and phenological shifts, significantly outperforming traditional machine learning classifiers like Random Forests [17]. However, standard CNNs lack an inherent mechanism to model strict, long-term sequential dependencies, as their receptive field is limited by the kernel size.

Recurrent Neural Networks (RNNs), GRUs and LSTMs

To explicitly model the chronological order of agricultural events, Recurrent Neural Networks (RNNs) became the standard. The foundational RNN Encoder-Decoder architecture allowed models to ingest variable-length sequences and maintain a hidden state that summarizes past inputs to inform future predictions [5]. To address the vanishing gradient problem inherent in basic RNNs, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) were widely adopted. These architectures are particularly advantageous in remote sensing, as specialized recurrent layers (such as GRU-D) can be engineered to handle the irregular time gaps, cloud occlusions, and missing values that commonly afflict satellite time series [36].

The Transformer Paradigm

Despite their success, LSTMs process data sequentially, which bottlenecks training parallelization and limits their ability to capture extremely long-range dependencies across a full growing season [37]. The introduction of Transformer architectures marked a paradigm shift in sequence modeling. By dispensing with recurrence and relying entirely on self-attention mechanisms, Transformers evaluate the relationships between all time steps simultaneously, regardless of their distance in the sequence [37]. In modern crop mapping applications, Transformers have demonstrated superior classification accuracy and generalizability compared to 1D-CNNs, especially when handling complex, multi-year, or irregularly sampled temporal features [27].

Hybrid Architectures

To maximize predictive performance, researchers have increasingly developed hybrid architectures that integrate multiple sequence-modeling paradigms. Early hybrid networks in remote sensing frequently combined CNNs with RNNs (such as ConvLSTMs or CNN+GRU architectures) to simultaneously extract spatial textures and temporal phenology [30]. As the focus has shifted toward capturing highly complex, multi-scale temporal dynamics, newer state-of-the-art approaches merge the localized sequential extraction capabilities of LSTMs with the global, long-range processing power of Transformers [11, 37].

While these standard hybrid models represent the current benchmark for decoding spatiotemporal Earth observation data, they typically process sequences within a single temporal scale. This thesis proposes a novel inversion of this paradigm specifically tailored for multi-year agricultural monitoring. Rather than using both networks on the same sequence, the proposed architecture separates the timescales: Transformers are utilized to concurrently extract dense, intra-year phenological features, while Bidirectional LSTMs are subsequently deployed to capture the long-term, inter-year sequential dependencies of crop rotations.

1.3.4 Multi-Year Crop Rotation Modeling

The Importance of Long-Term Agricultural Dynamics

Crop rotation is a fundamental agricultural practice wherein farmers alternate the species of crops grown on a specific plot across consecutive seasons. This strategy is essential for maintaining soil health, preventing nutrient depletion, and breaking pest and disease cycles. Despite its agronomic importance, most remote sensing models suffer from "temporal myopia," analyzing single-year phenology while ignoring inter-annual historical dependencies.

Traditional and Knowledge-Based Approaches

Early attempts to map multi-year dynamics relied on explicitly injecting prior agronomic knowledge into the classification pipeline. For example, probabilistic models such as Bayesian networks and Conditional Random Fields have been used to encode known crop rotation rules; the outputs of these networks are then fed into standard classifiers (such as Support Vector Machines) to guide the algorithm and improve early-season mapping accuracy [22, 30]. While effective, these methods require manually constructed rule bases and lack the flexibility to adapt to unexpected crop sequence changes.

Machine Learning and Temporal Transferability

As satellite archives have grown, researchers have increasingly utilized machine learning to track

land-use transitions over longer periods. Hierarchical classification workflows (such as tiered Random Forests) have been successfully deployed across multiple years to explicitly track crop rotations, field abandonment, and land-use intensity [3]. However, monitoring annual crop rotations over a decade presents significant challenges regarding temporal transferability; models trained on a specific year often struggle to generalize to past or future years due to shifting weather patterns, varying sensor availability, and inter-annual phenological shifts [27].

The Deep Learning Frontier

While early deep learning methods successfully tracked multi-year changes, they predominantly processed each year independently and relied on post-classification comparisons to identify rotations. The next frontier in precision agriculture mapping is the development of architectures that possess the intrinsic memory capacity to learn and predict multi-year agricultural cycles directly from continuous, raw satellite time-series data. Recently, researchers have demonstrated significant success in bridging these paradigms by integrating Bayesian modeling—such as Hidden Markov Models (HMMs)—with deep Transformer Encoders to explicitly capture intricate multi-year crop rotation patterns [26]. Building upon these foundations, the focus now shifts toward purely neural hybrid architectures capable of modeling these complex temporal dependencies end-to-end.

1.4 Contributions

This thesis advances the field of agricultural remote sensing by addressing the computational bottlenecks of large-scale mapping and exploring the predictive power of multi-year sequence modeling. The primary contributions of this work are as follows:

- **Adoption of an Optimized 1D Data Pipeline:** To overcome the limitations of memory-intensive 2D spatial patches, a streamlined, 1D pixel-based data representation is utilized. This approach drastically reduces High-Performance Computing (HPC) I/O bottlenecks, enabling the rapid processing and evaluation of massive, multi-year Satellite Image Time Series (SITS).
- **Comprehensive Baseline Benchmarking:** Robust single-year temporal baseline models, specifically a 1-Dimensional Convolutional Neural Network (1D-CNN) and a Gated Recurrent Unit Recurrent Neural Network (GRU-RNN), are developed, trained and thoroughly evaluated. These baselines provide a rigorous standard for intra-year phenological classification against which more complex models can be compared.
- **Novel Multi-Year Hybrid Architecture:** A custom Hierarchical Transformer-LSTM architecture is designed and implemented. This hybrid model uniquely combines the self-attention mechanisms of a Transformer with the recurrent memory of an LSTM to capture both intra-year weather irregularities and inter-year crop rotation dependencies.
- **Integration of Automated MLOps and Distributed Tuning:** A robust Machine Learning Operations (MLOps) workflow, leveraging PyTorch Lightning, Ray Tune, and Optuna, is adapted to automate the training pipeline. This infrastructure facilitates distributed hyperparameter optimization across an HPC cluster, ensuring that all architectures are evaluated fairly and at their maximum potential.

By delivering this end-to-end framework, this thesis not only provides a scalable software solution for operational crop mapping but also offers critical insights into whether neural networks can

effectively learn the historical, multi-year rules of agricultural crop rotation.

2 Methodology and System Design

2.1 The INVEKOS Dataset

To evaluate the proposed sequence-modeling architectures, experiments were conducted using the Austrian INVEKOS (Integrated Administration and Control System) dataset. INVEKOS serves as the European Union’s official spatial registry for agricultural parcel identification, providing rich, multi-year ground-truth vector data based on farmers’ declarations [26].

It is important to explicitly state that the data extraction and pre-processing pipeline utilized in this thesis was originally designed and implemented by Perantoni et al. [26]. By adopting their highly optimized dataset structure, this research was able to bypass standard preprocessing bottlenecks and focus entirely on the architectural development of neural network sequence models.

For this project, the spatial boundaries and crop labels were matched with Sentinel-2 Level-1C (L1C) optical imagery, specifically focusing on the Military Grid Reference System (MGRS) tile 33UVP [26]. This provided a continuous, six-year acquisition period spanning from September 2016 to September 2022, encompassing a total of 805 satellite acquisitions with all 13 spectral channels resampled to a 10m resolution [26].

Parameter	Value
Data Source	Sentinel-2 L1C & INVEKOS
Tile	MGRS 33UVP
Timeframe	Sept 2016 – Sept 2022
Total Acquisitions	805
Spatial Resolution	10m
Number of Classes	47
Minimum Parcel Area	400 m ²
Erosion Buffer	15m
Data Splits	60% Train, 20% Val, 20% Test

Table 2.1: Summary of the INVEKOS dataset and extraction parameters.

2.1.1 Data Acquisition (Spatial Patches vs. 1D Pixel Data)

A critical challenge in pixel-based crop mapping is the presence of ”mixed pixels” at the boundaries of agricultural parcels, which contain overlapping spectral signatures from neighboring fields. To mitigate this, a stringent geometric preprocessing pipeline was applied to the INVEKOS shapefiles [26].

Instead of extracting raw 2D spatial patches, spatial erosion buffers were applied to the interior of all field polygons to exclude border pixels. Furthermore, a minimum area threshold of 400 m² was enforced to ensure only sufficiently large and homogeneous parcels were retained [26]. The refined, eroded geometries were subsequently rasterized to align precisely with the Sentinel-2 pixel grid. Finally, to optimize memory overhead, each crop field was collapsed into a single 1D entry representing the spatial mean value of the pixels belonging to that field on every acquisition date [26]. This filtering process yielded an exceptionally clean dataset containing 47 distinct crop type labels [26].

2.1.2 Pre-processing and Handling Irregular Time Series

Handling massive spatial-temporal raster datasets alongside dense vector geometries introduces severe I/O bottlenecks in standard machine learning pipelines. To overcome this, the 1D Sentinel-2 time series data and corresponding ground-truth representations were converted into a streamlined HDF5

(.h5) format, and subsequently into a Lightning Memory-Mapped Database (LMDB) format.

This database architecture enables fast, memory-mapped data reading without requiring the entire multi-gigabyte dataset to be loaded into system RAM. As implemented through the project’s PyTorch Lightning `InvekosDataModule`, the dataset enforces a rigorous stratified random sampling partitioning strategy to ensure unbiased evaluation [26]. The data is split using a standard ratio of 60% for training, 20% for validation, and 20% for testing [26]. This optimized 1D data structure feeds perfectly into data loaders configured with multi-worker pre-fetching, ensuring high throughput across High-Performance Computing (HPC) nodes.

2.2 Single-Year Non-Hybrid Baselines

To establish a rigorous benchmark for multi-year architectures, two single-year sequence modeling baselines were developed and evaluated. These models process intra-year phenological data to provide a comparative standard against which the complexity of hybrid and multi-year models can be justified.

2.2.1 Temporal 1D Convolutional Neural Network (CNN)

The first baseline is a 1-Dimensional Convolutional Neural Network (1D-CNN) designed to extract temporal features by sliding convolutional filters across the time dimension, a methodology that has proven highly effective for satellite time series classification [24].

The **encoder** consists of two sequential convolutional blocks. The initial layer maps the 13 input satellite bands to 64 channels using a kernel size of 3 and a padding of 1 to preserve the temporal sequence length. This is followed by a 1D Batch Normalization layer and an in-place ReLU activation to introduce non-linearity. The second layer further projects the features from 64 to 128 channels, utilizing the same kernel size, padding, normalization, and activation parameters.

For **classification**, a point-wise convolution (kernel size of 1) is applied to map the extracted 128-dimensional features to the 47 crop classes. To generate the final field-level classification, the network executes a temporal majority vote by averaging the logit predictions across all time steps in the year.

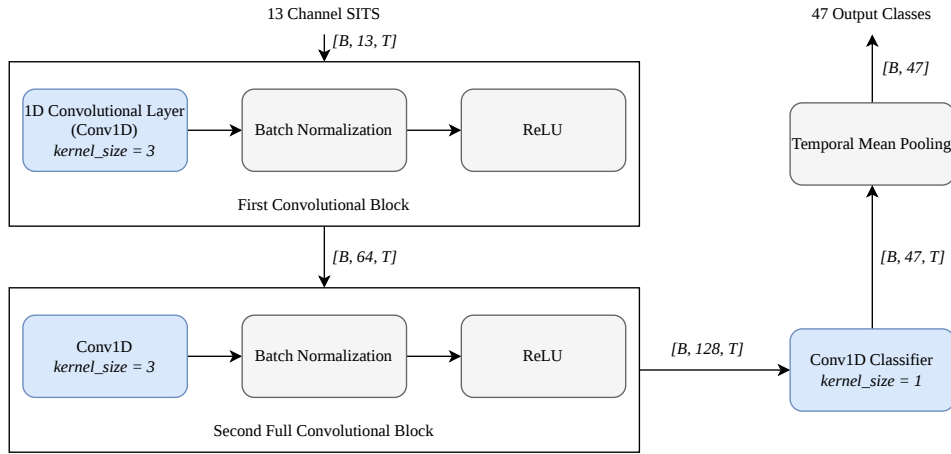


Figure 2.1: Architectural block diagram of the Temporal 1D-CNN baseline.

2.2.2 Temporal Gated Recurrent Unit (GRU RNN)

The second baseline leverages a Gated Recurrent Unit (GRU) [5] to explicitly model the sequential dependencies of agricultural phenology.

A **bidirectional GRU** architecture is utilized to process the temporal sequences in both forward and backward directions, yielding better performance than a standard unidirectional RNN. The input tensors are systematically permuted to align the time and channel dimensions for batch-first sequence processing. Instead of averaging predictions over time, the network isolates the final temporal output step. This final hidden state is fed into a fully connected linear **classifier** to predict the 47 crop classes. Because the network is bidirectional, the input dimension of the final linear classifier is

explicitly doubled to accommodate the concatenated forward and backward hidden features. To assess the optimal architectural capacity, the hyperparameter search space tests varying hidden dimensions (128, 256, and 512) and depth configurations (1, 2, and 3 stacked layers).

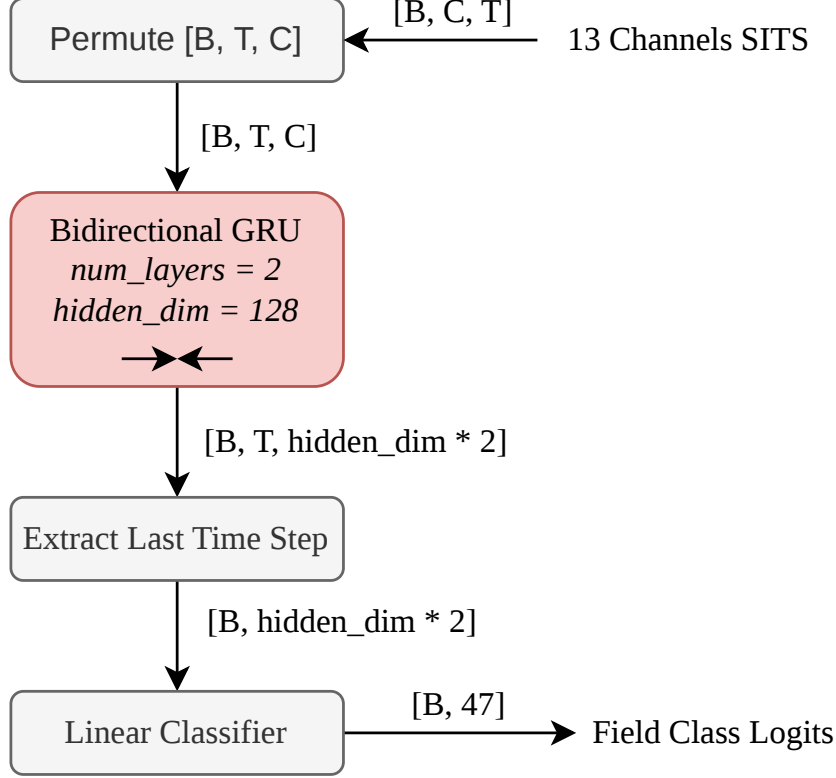


Figure 2.2: Architectural diagram of the Bidirectional GRU baseline.

2.3 Multi-Year Hybrid Architecture

The core contribution of this thesis is the development of a hybrid, multi-scale neural architecture capable of simultaneously modeling intra-year phenological dynamics and inter-year crop rotation patterns. Unlike traditional approaches that treat multi-year classification as independent yearly tasks, this architecture hierarchically processes continuous, multi-year sequences end-to-end.

2.3.1 AgroDays Positional Encoding

The first step in the data pipeline is preparing the raw inputs for sequence modeling. The raw 1D pixel sequence, consisting of 13 spectral channels, is initially projected into a higher-dimensional embedding space (128 dimensions). However, standard sequence models assume that data points arrive at regular, evenly spaced intervals. Agricultural satellite time series are inherently irregularly sampled due to cloud cover and varying satellite revisit times. Using standard sequential positional encodings would falsely inform the model that all observations are equidistant.

To address this, the architecture utilizes a custom **AgroDays Positional Encoding** module. Instead of simply indexing observations as sequential integer steps, this module adapts the standard sinusoidal positional encoding [33] by scaling the frequencies using the exact agricultural day of the acquisition.

Because sine and cosine are continuous periodic functions, mapping absolute day values onto these waves ensures that the mathematical distance between any two encoded time steps is proportional to

the actual temporal gap between them. By calculating the encoding angles based on this absolute temporal position, the model inherently understands whether consecutive satellite acquisitions are 5 days apart or 30 days apart. This explicit temporal injection ensures the network accurately aligns the spectral data with the correct stage of the crop’s phenological cycle.

Algorithm 1 AgroDays Positional Encoding

Require: Flattened input sequence $X \in \mathbb{R}^{(B \cdot Y) \times T \times d_{model}}$

Require: Agricultural days matrix $pos \in \mathbb{R}^{(B \cdot Y) \times T}$

Ensure: Position-encoded sequence $X_{enc} \in \mathbb{R}^{(B \cdot Y) \times T \times d_{model}}$

- 1: $div_term \leftarrow \exp \left(\text{arange}(0, d_{model}, 2) \cdot \frac{-\ln(10000.0)}{d_{model}} \right)$
 - 2: $pos_expanded \leftarrow \text{unsqueeze}(pos, \text{dim} = -1)$
 - 3: $angles \leftarrow pos_expanded \times div_term$
 - 4: $PE \leftarrow \text{zeros_like}(X)$
 - 5: $PE_{::,0::2} \leftarrow \sin(angles)$ ▷ Apply sine to even indices
 - 6: $PE_{::,1::2} \leftarrow \cos(angles)$ ▷ Apply cosine to odd indices
 - 7: $X_{enc} \leftarrow X + PE$ ▷ Inject temporal signal into embeddings
 - 8: **return** X_{enc}
-

As detailed in Algorithm 1, the implementation relies on several key intermediate variables to achieve this projection. The *div_term* represents the frequency scaling tensor, calculated logarithmically to ensure a geometric progression of wavelengths across the network’s core embedding dimension (*d_model*, defined as 128 in this architecture).

The input matrix *pos*, containing the absolute agricultural days, is broadcasted into *pos_expanded* and multiplied by the frequency scales to generate the *angles* tensor.

Finally, applying the sine and cosine operations to even and odd indices respectively yields the final positional encoding tensor, *PE*, which is added directly to the input sequence.

2.3.2 The Hierarchical Transformer-LSTM Model

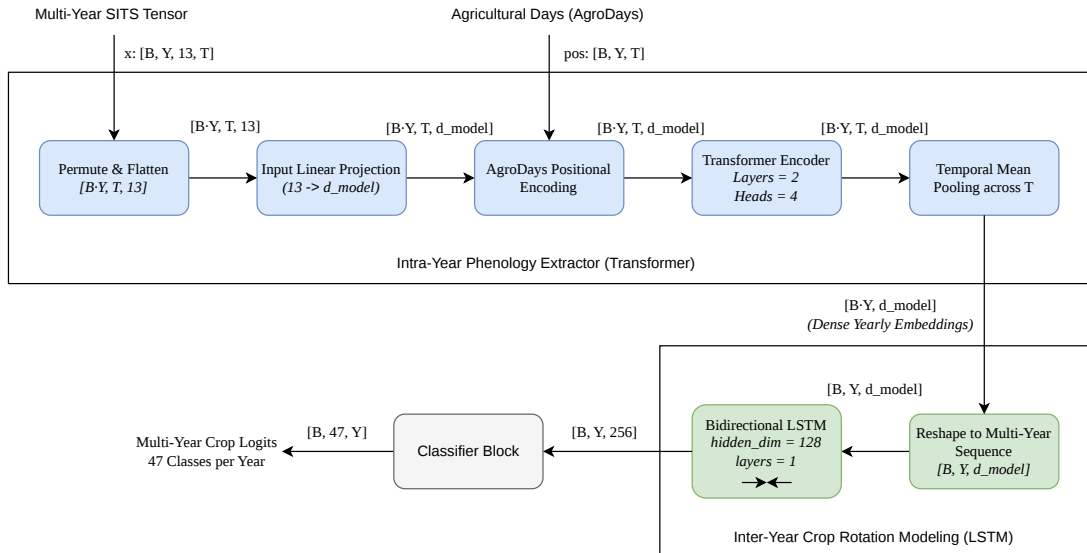


Figure 2.3: Hierarchical overview of the Transformer-LSTM architecture, demonstrating the fusion of intra-year and inter-year components.

Once the position-aware embeddings are generated, the network processes the temporal scales using a two-stage hierarchical model. In the first stage, the objective is to understand the intra-year phenological growth cycle.

Driven by Multi-Head Self-Attention [33], an Intra-Year Transformer Encoder evaluates the entire

yearly sequence simultaneously, effectively capturing complex growth stages without being hindered by vanishing gradients [29].

In the second stage, the architecture transitions from intra-year phenology to inter-year crop dependencies by passing the extracted features into an Inter-Year Long Short-Term Memory (LSTM) network [12].

Algorithm 2 Hierarchical Transformer-LSTM Forward Pass

Require: Multi-year input tensor $X \in \mathbb{R}^{B \times Y \times C \times T}$

Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times Y \times T}$

Ensure: Crop class logits $Y_{pred} \in \mathbb{R}^{B \times n_classes \times Y}$

```

1: Stage 1: Intra-Year Processing (Transformer)
2:  $X \leftarrow \text{permute}(X, [0, 1, 3, 2])$  ▷ Target shape:  $[B, Y, T, C]$ 
3:  $X_{flat} \leftarrow \text{reshape}(X, [B \cdot Y, T, C])$  ▷ Flatten for independent yearly processing
4:  $pos_{flat} \leftarrow \text{reshape}(pos, [B \cdot Y, T])$ 
5:  $X_{emb} \leftarrow \text{LinearProjection}(X_{flat})$ 
6:  $X_{emb} \leftarrow \text{AgroDaysPositionalEncoding}(X_{emb}, pos_{flat})$ 
7:  $T_{out} \leftarrow \text{TransformerEncoder}(X_{emb})$  ▷ Self-Attention over time steps
8:  $Y_{emb} \leftarrow \text{mean}(T_{out}, \text{dim} = 1)$  ▷ Temporal pooling:  $[B \cdot Y, d\_model]$ 
9: Stage 2: Inter-Year Processing (LSTM)
10:  $Y_{seq} \leftarrow \text{reshape}(Y_{emb}, [B, Y, d\_model])$  ▷ Restore sequence of years
11:  $L_{out}, (h_n, c_n) \leftarrow \text{BidirectionalLSTM}(Y_{seq})$  ▷ Model crop rotations
12: Stage 3: Classification
13:  $logits \leftarrow \text{LinearClassifier}(L_{out})$  ▷ Map to  $n\_classes$ 
14:  $Y_{pred} \leftarrow \text{permute}(logits, [0, 2, 1])$  ▷ Final shape:  $[B, n\_classes, Y]$ 
15: return  $Y_{pred}$ 

```

Algorithm 2 outlines the exact tensor transformations required to bridge these two temporal scales. To optimize hardware efficiency during the intra-year stage, the raw sequence X is flattened into X_{flat} , temporarily merging the batch (B) and year (Y) dimensions. This critical step allows the Transformer to process all years independently and in parallel across the GPUs.

The Transformer’s output sequence, T_{out} , is then temporally mean-pooled into Y_{emb} , collapsing the complex intra-year phenology into a single, dense feature vector for each specific year.

Finally, the chronological sequence is restored (Y_{seq}) and passed to the LSTM, which outputs the deeply fused multi-year features L_{out} alongside its final hidden and cell states (h_n, c_n).

2.3.3 Capturing Long-Term Crop Rotations

The primary motivation for integrating the LSTM in the second stage is its exceptional ability to model strict, long-term sequential rules. Traditional crop mapping architectures primarily focus on extracting phenological features within a single growing season, discarding a vast amount of predictive temporal context. However, agricultural practices are heavily governed by multi-year crop rotation cycles designed to maintain soil health and comply with policies.

To explicitly capture these long-term crop rotations, a bidirectional LSTM configuration is employed. This enables the network to analyze the rotation history in both the forward and backward temporal directions, effectively predicting a crop based on what was planted in previous years as well as what will be planted in subsequent years. By analyzing this bi-temporal history, the network learns deep inter-year transition probabilities.

Finally, the bidirectional output—representing this deeply fused, multi-year context—is passed to a fully-connected classification block. Because the LSTM is bidirectional, the hidden feature dimension is explicitly doubled to accommodate the concatenated forward and backward hidden states. These deep features are passed through a ReLU activation and a dropout regularization layer (set to 10%) to prevent overfitting. A final linear projection maps the processed features into the unnormalized logit scores for the 47 target crop classes, outputting a highly informed prediction for each year in the sequence.

2.4 Infrastructure and Optimization

Training deep temporal models on massive satellite datasets requires a robust, scalable infrastructure. To ensure reproducibility, efficient hardware utilization, and systematic hyperparameter selection, the experimental framework was constructed using PyTorch Lightning [8], Ray Tune [18], and MLflow [34].

2.4.1 PyTorch Lightning Training Pipeline

The training logic for all models was encapsulated within the PyTorch Lightning framework [8], abstracting code engineering while ensuring multi-GPU scalability. Several optimizations were implemented at the pipeline level to stabilize training and maximize throughput:

Mixed Precision Training: To optimize GPU memory and accelerate tensor operations, the trainer was configured to use 16-bit mixed precision (`16-mixed`). This allows the models to perform computationally heavy operations in half-precision while maintaining full precision for gradient accumulations, maximizing the throughput of the underlying hardware.

Loss Formulation: The networks were trained using a standard Cross-Entropy Loss function. Crucially, an `ignore_index` parameter set to 255 was passed to the loss function. This ensured that any padded sequences or years dropped during data augmentation did not contribute to the loss gradients or skew the accuracy metrics.

Optimization and Regularization: Optimization was governed by the AdamW algorithm [19]. Unlike the standard Adam optimizer, AdamW correctly decouples weight decay from the adaptive learning rate updates, providing superior regularization [19]. To prevent unnecessary computational expenditure on converging models, an Early Stopping callback was implemented, terminating training if the validation accuracy (`val_acc`) failed to improve for 10 consecutive epochs.

2.4.2 Distributed Hyperparameter Tuning with Ray Tune and Optuna

Neural network sequence models are highly sensitive to their hyperparameters. To systematically search for the optimal configuration, a distributed tuning loop was orchestrated using Ray Tune [18], integrated with the Tree-structured Parzen Estimator (TPE) sampling algorithm from Optuna [2].

The tuning process was executed with a budget of 30 total trials. The Optuna sampler was initialized with 20 completely random startup trials to thoroughly explore the boundaries of the parameter space before transitioning to a Bayesian optimization strategy. To maximize hardware efficiency, Ray Tune was configured to run 3 trials concurrently. During this exploratory phase, each trial was trained for a truncated duration of 5 epochs. The continuous search space included a log-uniform distribution for the learning rate (bounded between $1e-5$ and $1e-2$) and the weight decay (bounded between $1e-7$ and $1e-3$). Discrete grid-search choices were provided for the batch size (8, 16, or 32). Furthermore, architecture-specific parameters were dynamically tuned:

- For the **GRU RNN**, the search space tested hidden dimensions of 128, 256, and 512, alongside variable network depths of 1, 2, or 3 stacked layers.
- For the **Transformer-LSTM**, the core embedding dimension (`d_model`) was tested across 128, 256, and 512.

Every trial automatically logged its hyperparameters and performance metrics (including training loss, validation loss, overall accuracy, and F1-score) to a local MLflow database [34]. After the tuning phase concluded, the configuration that yielded the maximum validation accuracy was extracted. This optimal configuration was then used to instantiate a final model, which was trained from scratch for 40 epochs and rigorously evaluated on the held-out test set.

Hyperparameter	Distribution/Values	Applies To
Learning Rate	Log-Uniform [1e−5, 1e−2]	All Models
Weight Decay	Log-Uniform [1e−7, 1e−3]	All Models
Batch Size	{8, 16, 32}	All Models
GRU Hidden Dim	{128, 256, 512}	GRU RNN
GRU Layers	{1, 2, 3}	GRU RNN
Transformer <i>d_model</i>	{128, 256, 512}	Transformer-LSTM

Table 2.2: Hyperparameter search space defined for Ray Tune optimization.

3 Experiments and Discussion

3.1 Experimental Setup

To rigorously evaluate the proposed multi-year architecture against single-year baselines, a standardized experimental framework was established. All models were trained, validated, and tested using identical data splits and evaluation metrics to ensure a fair comparative analysis.

3.1.1 HPC Environment and I/O Bottleneck Mitigation

Training deep sequence models on massive, high-dimensional satellite time series is computationally demanding. To facilitate this, all experiments were executed on a dedicated High-Performance Computing (HPC) node equipped with an AMD Ryzen 7 5800X processor, 67GB of system RAM, and an NVIDIA GeForce RTX 3060 GPU, running on an Ubuntu 22.04 LTS environment.

A primary challenge in processing the INVEKOS dataset is the severe input/output (I/O) bottleneck caused by loading thousands of multi-year temporal sequences. Initial exploratory experiments attempted to utilize the full 2D spatial patches of the agricultural parcels. However, loading continuous multi-year, multi-spectral spatial patches introduced an insurmountable I/O bottleneck. Because the 2D patch dataset was exceptionally large (approximately 1.3 TB), it could not be stored on the HPC node’s local solid-state drive (SSD) and instead had to be hosted on a shared network folder. The immense volume of network-based disk reads required to process the 2D geometries caused severe GPU starvation, rendering the training process computationally infeasible despite the HPC infrastructure.

To resolve this, the experimental framework was adapted to utilize the highly optimized 1D pixel LMDB dataset provided by Perantoni et al. [26]. As detailed in the methodology, adopting this memory-mapped structure effectively eliminated the I/O bottleneck. It allowed the PyTorch Lightning data loaders to rapidly access sequences directly from disk, sustaining high GPU utilization without the latency overhead of the 2D spatial patches. Furthermore, all model executions, hyperparameter tuning trials via Ray Tune, and resulting performance metrics were systematically tracked and versioned using a centralized MLflow server.

3.1.2 Data Loading Strategies

To properly evaluate the impact of multi-year memory, the experimental framework utilized two distinct data loading strategies tailored to the architectural needs of the models. For the single-year baseline models (1D-CNN and Bidirectional GRU), the dataset was structured using an intra-year loader. This loader effectively decoupled the temporal sequence of the dataset, extracting and serving only a single growing season (e.g., 2018) at a time. This constrained the baselines to rely entirely on localized, intra-year phenological signatures for classification.

In contrast, the proposed Hybrid Transformer-LSTM required a full sequence view of the crop rotation cycle. To facilitate this, an inter-year data loader was engineered to serve the entire continuous

multi-year sequence for a given agricultural parcel simultaneously. When passed to the Hybrid model, this multi-year tensor is first processed by the Intra-Year Transformer, which concurrently analyzes the phenology within each individual year. The summarized yearly representations are then passed to the Inter-Year LSTM to sequentially process the multi-year crop rotation. This explicit separation in data handling guarantees that the performance delta observed between the architectures is strictly a product of the Hybrid model’s multi-year memory capabilities.

3.1.3 Evaluation Metrics

Agricultural land cover datasets are inherently unbalanced; a few dominant crop types (such as maize or winter wheat) often account for the vast majority of cultivated land, while specialized crops represent a tiny fraction of the dataset. Consequently, relying solely on overall accuracy can yield highly misleading evaluations. To provide a comprehensive assessment, the models were evaluated using two primary metrics: **Overall Accuracy** and the **Macro F1-Score**.

Overall Accuracy (OA) measures the total proportion of correctly classified agricultural parcels across all available classes. It is defined as:

$$OA = \frac{\sum_{i=1}^C TP_i}{N}$$

where C is the total number of target classes (47 in this study), TP_i represents the True Positives for class i , and N is the total number of samples in the dataset.

Macro F1-Score is the primary metric used to evaluate true model performance in this study. The F1-score for a single class is the harmonic mean of Precision and Recall. To compute the Macro F1-score, the F1-score is calculated independently for each of the 47 classes and then unweightedly averaged:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^C \left(2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \right)$$

Unlike a weighted F1-score, the Macro F1-score treats every class equally, regardless of its support size. A model can only achieve a high Macro F1-score if it successfully learns to identify both the dominant and the minority crop classes, making it the most rigorous benchmark for this highly imbalanced dataset.

3.2 Baseline Results (Non-Hybrid Models)

To quantify the value of modeling multi-year crop rotations, the performance of the proposed architecture must first be contextualized against established single-year models. To ensure statistical robustness and account for weight initialization variances, all baseline models were trained and evaluated across three distinct seeds.

3.2.1 Temporal CNN Performance

Following the hyperparameter tuning phase, the optimal Temporal 1D-CNN was configured with a batch size of 16, a learning rate of $2.66\text{e-}3$, and a weight decay of $1.65\text{e-}6$. The model was trained for 40 epochs.

As shown in Table 3.1, the CNN achieved a mean overall test accuracy of $55.17 \pm 0.00\%$. However, the model struggled significantly with the class imbalance of the INVEKOS dataset, recording a mean test Macro F1-score of only $26.78 \pm 0.00\%$. The large discrepancy between the overall accuracy and the F1-score indicates that while the CNN can adequately identify majority crop classes based on local temporal convolutions, it fails to capture the defining phenological features of minority classes.

3.2.2 Temporal GRU RNN Performance

The Bidirectional GRU baseline, designed to explicitly model sequential dependencies, was trained under similar conditions. The optimal configuration utilized a hidden dimension of 128 and a network depth of 3 stacked layers. The network was trained with a learning rate of $4.73\text{e-}4$, a weight decay of $4.98\text{e-}6$, and a batch size of 16.

Across the three random seeds, the GRU achieved a mean overall test accuracy of $57.78 \pm 0.00\%$

and a mean Macro F1-score of $34.95 \pm 0.00\%$. Compared to the local sliding window of the CNN, the recurrent memory of the GRU provided a measurable improvement in distinguishing the complex, irregular time gaps inherent in the Sentinel-2 acquisitions, raising the F1-score by approximately 8.17%. Ultimately, however, both single-year baselines demonstrate a severe performance ceiling when isolated from historical crop rotation data.

3.3 Proposed Architecture Results (Hybrid Model)

3.3.1 Hierarchical Transformer-LSTM Performance

To evaluate the core hypothesis of this thesis—that integrating inter-year crop rotation memory improves classification over isolated single-year models—the Hierarchical Transformer-LSTM was evaluated under an identical experimental setup. Following the Bayesian hyperparameter search, the optimal multi-year architecture was configured with a Transformer embedding dimension (d_{model}) of 256, utilizing 2 encoder layers and 4 attention heads. The inter-year sequence was processed by a single-layer Bidirectional LSTM with a hidden dimension of 128. The model was trained using a batch size of 16, a learning rate of $1.57e-4$, a weight decay of $2.91e-5$, and a dropout rate of 10% to prevent overfitting.

Consistent with the baselines, the optimal configuration was trained across three distinct random seeds. As detailed in Table 3.1, the hybrid architecture achieved a remarkable mean overall test accuracy of $80.65 \pm 0.51\%$. More importantly, it achieved a mean Macro F1-score of $61.17 \pm 0.50\%$, vastly outperforming both single-year baselines.

This dramatic F1-score improvement of over 26% compared to the GRU confirms that multi-year memory enables the network to correctly identify minority crop classes that were previously completely misclassified by the CNN and GRU. By understanding what was planted in the same field in previous years, the LSTM component effectively narrows down the probability distribution of current-year crops, breaking through the performance ceiling of intra-year models.

Model	Test Accuracy (%)	Test Macro F1 (%)	Test Loss
Temporal 1D-CNN (Single-Year)	55.17 ± 0.00	26.78 ± 0.00	1.46 ± 0.00
Bidirectional GRU (Single-Year)	57.78 ± 0.00	34.95 ± 0.00	1.40 ± 0.00
Transformer-LSTM (Multi-Year)	80.65 ± 0.51	61.17 ± 0.50	0.61 ± 0.01

Table 3.1: Final performance comparison across all evaluated architectures. Results are averaged across 3 random seeds, displaying the mean and standard deviation.

3.4 Discussion

The experimental results validate the central premise of this research: relying solely on intra-year phenology severely limits crop classification performance, particularly on highly imbalanced datasets. The hierarchical multi-year architecture not only surpassed the performance ceiling of the baselines but also demonstrated unexpected computational efficiencies.

3.4.1 Single-Year vs. Multi-Year Capabilities

The Hybrid Transformer-LSTM significantly outperforms the single-year baselines. While the Bidirectional GRU is theoretically better suited to capture complex temporal phenology than the CNN, it only achieved a Macro F1-score of 34.95%. In contrast, the Hybrid model reached a mean F1-score of 61.17%. This massive improvement highlights a fundamental limitation of single-year models: they evaluate each growing season in a vacuum. In agricultural reality, planting decisions are heavily governed by strict multi-year crop rotation practices designed to prevent soil depletion (e.g., alternating nitrogen-fixing legumes with demanding crops like maize).

By feeding a sequence of yearly embeddings into the inter-year LSTM, the Hybrid model learns these transitional probabilities. If the network is uncertain about a crop’s spectral signature in year t , it can infer the likely crop by analyzing the confident predictions from years $t-1$ and $t+1$. Furthermore,

despite its architectural complexity, the Hybrid model proved to be more computationally efficient during training, completing its required epochs across all three seeds faster than the single-year models.

This efficiency can likely be attributed to the model’s hierarchical design. As originally established by Vaswani et al. [33], self-attention mechanisms allow for significantly more parallelization than recurrent networks, which are bottlenecked by step-by-step sequential processing. While the baseline GRU must process hundreds of individual satellite acquisitions sequentially, the Hybrid model leverages the Intra-Year Transformer to evaluate the intra-year sequences concurrently. By pooling these outputs into dense, single-year vectors, the sequential computation burden on the inter-year LSTM is reduced to a sequence length of merely six steps (representing the six years of data).

3.4.2 The Impact of Class Imbalance

The INVEKOS dataset, like most real-world agricultural registries, suffers from severe long-tail class imbalance, dominated by staple crops such as winter wheat and maize [26]. This imbalance is fully exposed by the baseline models. The 1D-CNN achieved an overall accuracy of 55.17%, but an F1-score of only 26.78%. This nearly 30% discrepancy indicates a "lazy" network: as is common in highly imbalanced datasets, the model minimizes its training loss by over-predicting the majority classes and effectively ignoring the minority crops [13].

The Hybrid model significantly mitigates this failure mode. While its overall accuracy rose to 80.65%, its F1-score jumped disproportionately higher to 61.17%. This proves that the multi-year memory explicitly helps rescue minority classes. Rare or specialized crops are often planted as transitional phases within a longer crop rotation cycle to manage soil health [4, 22]. By recognizing the complete multi-year cycle, the LSTM is forced to predict the correct minority class to satisfy the learned rotational sequence, preventing the model from defaulting to the statistically safe, majority-class prediction.

Ultimately, these findings demonstrate that treating crop mapping as an isolated, single-year computer vision problem discards critical agronomic context. Fusing intra-year attention with inter-year recurrence is essential for accurate, scalable, and computationally viable agricultural monitoring.

4 Conclusions and Future Work

4.1 Conclusions

4.2 Future Work

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